THE REMIX

Covariates in DiD

Day 2 of 5: Afternoon

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The PT-violation problem, take two

- This morning: a **third group** that shares the differential trend.
- But often, you don't have a clean placebo group.
- This afternoon: **condition on covariates that absorb the differential trend.**

Today's three estimators, and one false friend

OR (outcome regression) **IPW** (inverse propensity weighting) **DR** (doubly robust)

And: "*can't I just control for the propensity score in a regression?*"
That's the false friend. We'll see why before DR.

The applied problem

- Your treated states differ from your controls on observables — baseline income, demographics, industry mix, education.
- If $Y(0)$ trends differ across these dimensions, unconditional PT is implausible.
- But maybe *within* strata of X, PT holds:

$$\mathbb{E}[\Delta Y(0) \mid D = 1, X] \;=\; \mathbb{E}[\Delta Y(0) \mid D = 0, X]$$

- This is **conditional parallel trends**. The escape valve for designs without a placebo group.
- Three estimators target this. They differ in *which side* of the conditioning they model.

This afternoon, six beats

1. Outcome regression (Heckman-Ichimura-Todd, 1997) — the bias OLS-with-controls actually identifies, plus the conditional-PT assumption that anchors the rest of the afternoon
2. Inverse propensity weighting (Abadie 2005)
3. Doubly robust DiD (Sant'Anna & Zhao 2020)
4. Hidden linearity bias — TWFE-with-controls fully exposed (Caetano & Callaway 2024)
5. Compositional change (Sant'Anna & Xu)
6. Choosing & checking X — the practical close

We lead with the bias of OLS. Each fix below removes one piece of that bias.

LESSON

1. Outcome Regression

What OLS-with-controls gives you — and what it doesn't

Conditional parallel trends, formally

The new identifying assumption

$$\mathbb{E}[\Delta Y(0) \mid D{=}1, X] \;=\; \mathbb{E}[\Delta Y(0) \mid D{=}0, X]$$

- Within strata of X, the untreated trend is the same for treated and control.
- Equivalently: *any* differential trend across treated and control is fully explained by their differences in X.
- Trade-off: weaker than unconditional PT (good), but you have to pick the right X (hard).
- Two ways to use X: model the outcome side ($\mu(X)$), or model the treatment side ($p(X)$). Each is one of our estimators.

A note on history of thought

The first articulation of any parallel-trends-type assumption in the DiD literature is:

> **Heckman, Ichimura & Todd (1997), *Review of Economic Studies***
>
> "Matching as an econometric evaluation estimator: evidence from evaluating a job training programme."

The irony. What HIT wrote down was the ***conditional*** PT assumption — $\mathbb{E}[\Delta Y(0) \mid D{=}1, X] = \mathbb{E}[\Delta Y(0) \mid D{=}0, X]$.

Unconditional PT came later. The morning's version — $\mathbb{E}[\Delta Y(0) \mid D{=}1] = \mathbb{E}[\Delta Y(0) \mid D{=}0]$ — is the special case of HIT's assumption with no X.

The first PTA written down was the conditional one. The unconditional one is what the textbook taught next.

Before any fix: what does OLS-with-controls actually identify?

The workhorse: $Y_{it} = \alpha_i + \gamma_t + \alpha\, D_{it} + \beta\, X_{it} + \varepsilon_{it}$. Estimate $\widehat{\alpha}$ by TWFE-OLS.

The Caetano-Callaway (2024) decomposition

$$\widehat{\alpha}_{\text{TWFE+}X} \xrightarrow{p} \underbrace{\overline{\text{ATT}}_w}_{\text{weighted ATT}} + \text{Bias}_A + \text{Bias}_B + \text{Bias}_C$$

- It is *not* the ATT. It is a *weighted average* of conditional ATTs — possibly with perverse weights — **plus** three bias terms.
- Each bias is a different assumption silently being made.
- **The bias is what creates the demand for OR, IPW, and DR.** They are each fixes for a piece of this expression.

Bias A: heterogeneous τ in X

In potential outcomes:

$$\tau(X) \;\equiv\; \mathbb{E}[Y(1) - Y(0) \mid X,\, D{=}1] \;\; \text{is a } \textit{function} \text{ of } X.$$

The true ATT averages $\tau(X)$ over the treated X-distribution:

$$\text{ATT} \;=\; \mathbb{E}\big[\tau(X) \mid D{=}1\big].$$

What TWFE-additive-X fits: one scalar $\hat{\delta}$ — a weighted average of $\tau(X)$ whose weights depend on OLS mechanics (the variance of D given X), *not* the treated X-distribution.

In plain English. Treatment works better for some types of people than others (e.g., training helps younger workers more). TWFE fits one number for everyone — a different weighted average than the one you wanted.

Bias B: X moves differently over time across groups

For PT-on-Y to hold under the TWFE-additive-X model:

$$\mathbb{E}[\Delta Y(0) \mid D{=}1] \;=\; \mathbb{E}[\Delta Y(0) \mid D{=}0].$$

Under the model $Y(0)_{it} = \alpha_i + \gamma_t + \beta X_{it} + \varepsilon$, this requires:

$$\beta \cdot \big(\mathbb{E}[\Delta X \mid D{=}1] - \mathbb{E}[\Delta X \mid D{=}0]\big) \;=\; 0.$$

Two ways for this to hold: $\beta = 0$ (the covariate is irrelevant), or $\Delta X^T = \Delta X^C$ (X trends the same way across groups). Otherwise: bias.

In plain English. The covariate trends differently for treated and control — say, the policy nudged X for one group but not the other — and because Y depends on X, that group difference shows up in the DiD as a fake treatment effect.

Bias C: the FE transformation eats time-invariant covariates

Within-unit demeaning: $\tilde{X}_{it} = X_{it} - \bar{X}_i$. If X is time-invariant for unit i (race, education at baseline, pre-program earnings), then $\tilde{X}_{it} = 0$ in every period.

$$\text{TWFE-additive-}X \quad \text{estimates as if those covariates were} \quad 0.$$

The same goes for any time-invariant Z. *What you wrote as a control isn't what the regression conditioned on.*

The fix: Z has to enter as $(\text{Post} \times Z)$ — which the FE cannot eat, because post varies within unit.

In plain English. You thought you controlled for race in LaLonde. The regression didn't see race — it demeaned it to zero. That's why additive-X on LaLonde gives the same answer as no covariates at all.

Outcome regression: model the trend, impute the counterfactual

Heckman, Ichimura & Todd (1997).

1. Use *only the control group* to fit a model of ΔY on X:

$$\widehat{\mu}_0(X) \;=\; \widehat{\mathbb{E}}[\Delta Y \mid D{=}0, X]$$

2. For each treated unit, predict the counterfactual trend $\widehat{\mu}_0(X_i)$.
3. The OR-DiD estimator:

$$\widehat{\delta}_{\text{OR}} \;=\; \frac{1}{N_1} \sum_{i:\, D_i=1} \big(\Delta Y_i \;-\; \widehat{\mu}_0(X_i)\big)$$

Interpretation: OR *imputes* what the treated would have done if untreated — using only the control group's X-to-trend mapping.

The TWFE-with-controls regression — and the three assumptions it hides

$$Y_{it} \;=\; \alpha_i \;+\; \gamma_t \;+\; \delta\,(T_t \times D_i) \;+\; \beta\, X_{it} \;+\; \varepsilon_{it}$$

- Standard move: add X_{it} to the TWFE. *“Sure! If you’re willing to impose three more assumptions.”*
- **(1) Additive functional form** in X (linear, no $D \times X$ interaction).
- **(2) Homogeneous treatment effects** in X (the slope β doesn’t depend on D).
- **(3) No X-specific trends differing across treated and control.**

Next slide: the proof of (2). Where does the homogeneity-in-X assumption come from? Trace it through the conditional expectations.

Proof, step 1 — the TWFE model and what conditional PT gives us

The TWFE model with one covariate X:

$$Y_{it} \;=\; \alpha_1 + \alpha_2\, T_t + \alpha_3\, D_i + \delta\,(T_t \times D_i) + \beta\, X_{it} + \varepsilon_{it}.$$

Conditional parallel trends says, for each value of X:

$$\mathbb{E}[Y_1^0 - Y_0^0 \mid D{=}1, X] \;=\; \mathbb{E}[Y_1^0 - Y_0^0 \mid D{=}0, X].$$

Rearrange. The missing counterfactual for the treated is identified:

$$\mathbb{E}[Y_1^0 \mid D{=}1, X] \;=\; \mathbb{E}[Y_0 \mid D{=}1, X] + \mathbb{E}[Y_1 \mid D{=}0, X] - \mathbb{E}[Y_0 \mid D{=}0, X].$$

Three observed conditional means $\Rightarrow$ one unobserved counterfactual. That's all PT does for us.

Proof, step 2 — plug TWFE in, the slopes look the same

From the TWFE model, the observed conditional means are:

$$\mathbb{E}[Y_0 \mid D{=}1, X] = \alpha_1 + \alpha_3 + \beta\, X$$
$$\mathbb{E}[Y_1 \mid D{=}0, X] = \alpha_1 + \alpha_2 + \beta\, X$$
$$\mathbb{E}[Y_0 \mid D{=}0, X] = \alpha_1 + \beta\, X$$

Plug into the identity from step 1:

$$\mathbb{E}[Y_1^0 \mid D{=}1, X] \;=\; \alpha_1 + \alpha_2 + \alpha_3 + \beta\, X.$$

The treated Y^1 post-period, from the TWFE model:

$$\mathbb{E}[Y_1^1 \mid D{=}1, X] \;=\; \alpha_1 + \alpha_2 + \alpha_3 + \delta + \beta\, X.$$

Both potential outcomes carry the same β in X. Subtract: ATT $= \delta$. Clean. Suspiciously clean.

Proof, step 3 — what if Y^1 and Y^0 have different slopes in X?

Let the slopes differ. Write β_1 for the Y^1 slope and β_2 for the Y^0 slope:

$$\begin{aligned} \mathbb{E}[Y_1^1 \mid D{=}1, X] &= \alpha_1 + \alpha_2 + \alpha_3 + \delta + \boxed{\beta_1}\, X \\ \mathbb{E}[Y_1^0 \mid D{=}1, X] &= \alpha_1 + \alpha_2 + \alpha_3 \qquad + \boxed{\beta_2}\, X \end{aligned}$$

The true ATT given X:

$$\text{ATT}(X) \;=\; \mathbb{E}[Y^1 - Y^0 \mid D{=}1, X] \;=\; \delta \;+\; (\beta_1 - \beta_2)\, X.$$

- If $\beta_1 \neq \beta_2$, the treatment effect *depends on* X.
- TWFE fits **one** β, not two.
- So $\widehat{\delta}$ recovers the ATT only when $\beta_1 = \beta_2$.

Proof, step 4 — Assumption 4: homogeneous τ in X

Assumption 4 (Homogeneous treatment effects in X)

TWFE-with-X requires the treatment effect to be the same at every value of X.

- If X is **sex**: the effect is the same for men and for women.
- If X is **income** (continuous): the effect is the same at \$1 and at \$1,000,000.
- If X is **age**: the effect is the same at 25 and at 65.

This is a structural assumption on the DGP. *OLS doesn't tell you you're making it. The math does.*

If the effect actually varies with X, TWFE-with-X averages across the variation and gives you a weighted parameter you didn't ask for.

Proof, step 5 — the goal: parallel trends on Y^0, and TWFE's model

What we need. Parallel trends on the untreated potential outcome:

$$\mathbb{E}[Y^0_{\text{post}} - Y^0_{\text{pre}} \mid D{=}1] \;=\; \mathbb{E}[Y^0_{\text{post}} - Y^0_{\text{pre}} \mid D{=}0]$$

TWFE's structural model for Y^0:

$$Y^0_{it} \;=\; \alpha_1 \;+\; \alpha_2\, T_t \;+\; \alpha_3\, D_i \;+\; \beta\, X_{it} \;+\; \varepsilon^0_{it}$$

β is the slope of Y^0 on X. The question for the next slides: what does the model assume to make PT hold?

Proof, step 6 — Y^0's within-group trend, by subtraction

For group D, write the post and pre conditional means:

$$\mathbb{E}[Y^0_{\text{post}} \mid D] \;=\; \alpha_1 + \alpha_2 + \alpha_3 D + \beta\, \mathbb{E}[X \mid D, \text{post}]$$
$$\mathbb{E}[Y^0_{\text{pre}} \mid D] \;=\; \alpha_1 \qquad\quad + \alpha_3 D + \beta\, \mathbb{E}[X \mid D, \text{pre}]$$

Subtract: the α_1 and $\alpha_3 D$ cancel.

$$\boxed{\mathbb{E}[Y^0_{\text{post}} - Y^0_{\text{pre}} \mid D] \;=\; \alpha_2 \;+\; \beta \cdot \Delta X^D}$$

where $\Delta X^D \equiv \mathbb{E}[X|D, \text{post}] - \mathbb{E}[X|D, \text{pre}]$.

Proof, step 7 — PT forces $\beta\,(\Delta X^T - \Delta X^C) = 0$

Set the two trends equal (PT on Y^0):

$$\alpha_2 + \beta\,\Delta X^T \;=\; \alpha_2 + \beta\,\Delta X^C$$

The α_2 cancels. Rearrange:

$$\boxed{\beta\,(\Delta X^T - \Delta X^C) \;=\; 0}$$

Two ways for this to hold:

- $\beta = 0 \;\Rightarrow\; Y^0$ doesn't depend on X.
- $\Delta X^T = \Delta X^C \;\Rightarrow\; X$-trends parallel across groups.

Proof, step 8 — Assumptions 5 & 6: Y^0 trends don't depend on X

Assumptions 5 & 6

Y^0's trend over time does not depend on X — within either the treated or the control group.

- Rich people cannot trend differently than poor people.
- Men cannot trend differently than women.
- College grads cannot trend differently than high-school dropouts.

The realized-X version ("X-trends equal") is a consequence, not the assumption.

Without these six (1–3 baseline DiD plus 4, 5, 6), TWFE-with-X does not identify the ATT.

Four ways to break TWFE-with-X

Generate $Y(0)$ *directly*, cell-by-cell — so the true DGP isn't $\alpha + \delta D + \beta X + \varepsilon$. Each version turns on one knob.

- **DGP 0 — clean linear.** $Y(0) = \alpha_i + \gamma_t + \beta X + \varepsilon$. No violation. Sanity check.
- **DGP 1 — time-varying β.** $\beta_{\text{pre}} = 1,\ \beta_{\text{post}} = 3$. X's effect on Y shifts at the policy break.
- **DGP 2 — heterogeneous τ in X.** $\tau_i = 2 + 4(X_i - \frac{1}{2})$. Treatment effect grows with X.
- **DGP 3 — time-invariant Z.** $Y(0)$ depends on a baseline Z only in post. FE within-unit transformation *eats* Z.

True $\tau = 2$ in every DGP. Selection: treated units' X shifts up by 0.3 in post (the typical empirical pattern).

The DGP, in R

```r
# Common setup
N <- 4000
units <- data.frame(id = 1:N,
                    D = rbinom(N, 1, 0.5),
                    Z = rbinom(N, 1, 0.5))
panel <- merge(units, data.frame(t = c(0, 1)))
panel$post <- as.integer(panel$t == 1)
panel$X <- runif(nrow(panel)) + 0.3 * panel$D * panel$post

# Y(0), one of four versions:
panel$Y0 <- panel$alpha + 0.5*panel$post + 1*panel$X + panel$eps # DGP 0
# Y0 = ... + ifelse(panel$post == 0, 1, 3) * panel$X + ... # DGP 1
# tau = 2 + 4*(X - 0.5) # DGP 2
# Y0 = ... + 5*panel$Z*panel$post + ... # DGP 3

panel$Y <- panel$Y0 + panel$D * panel$post * panel$tau
```

Full script: `decks/code/Covariates-Bias/twfe_bias_dgp.R`.

Results: which spec fixes which bias? (true $\tau = 2$)

	no covariates	**additive X**	**Post$\times X$**	**Post$\times X + Z$**
DGP 0 clean	2.30	2.02	2.01	2.01
DGP 1 time-varying β	2.91	2.30	1.99	1.99
DGP 2 het τ in X	3.51	2.91	2.64	2.64
DGP 3 time-inv Z	2.15	1.85	1.83	1.98

green: spec recovers $\tau \approx 2$. *red: spec is biased.*

The bias-to-fix map

- **Time-varying** β $\rightarrow$ the fix is Post$\times X$. Let X's effect move with the policy break.
- **Heterogeneous** τ **in** X $\rightarrow$ Post$\times X$ *still misses it.* You need a saturated $D \times X$ interaction, or DR-DiD (Lesson 4), or matching.
- **Time-invariant** Z $\rightarrow$ the fix is Post$\times Z$. Additive Z does nothing because FE absorbs it.

The unifying lesson (Caetano & Callaway 2024)

Additive-X TWFE is one regression that silently assumes *all three* of these don't apply. When they do, the bias has a name: **hidden linearity bias.** Each fix is a relaxation.

The proper TWFE spec — and what it still can't handle

To kill Bias B and Bias C inside TWFE, you need:

$$Y_{it} \;=\; \alpha_i + \gamma_t + \delta\,(D_i{\times}\text{post}_t) \;+\; X_{it}'\gamma_X \;+\; (\text{post}_t \cdot X_{it})'\gamma_{pX} \;+\; (\text{post}_t \cdot Z_i)'\gamma_{pZ} \;+\; \varepsilon_{it}$$

- $(\text{post} \times X)$ handles Bias B: lets X's effect shift at the policy break.
- $(\text{post} \times Z)$ handles Bias C: rescues time-invariant baseline characteristics from the FE.

What this spec *cannot* fix: Bias A. If τ depends on X, no single $\hat{\delta}$ recovers the ATT.

The next move: estimators that handle Bias A by averaging over the treated X-distribution explicitly.

Three doors to the same fix

Three estimators that handle heterogeneous $\tau(X)$ — each does it differently:

- **Outcome regression (OR), Heckman, Ichimura & Todd (1997).**
  Fit $\widehat{\mu}_0(X)$ on controls. Impute the counterfactual for each treated unit. Average over the treated X-distribution. *This lesson.*
- **Inverse propensity weighting (IPW), Abadie (2005).**
  Reweight the controls by $\widehat{p}(X)/(1-\widehat{p}(X))$ to match the treated X-distribution. *Lesson 2.*
- **Doubly-robust DiD, Sant'Anna & Zhao (2020).**
  Combine OR and IPW. Consistent if *either* model is right. *Lesson 3.*

Three doors, same room. The choice between them is mostly a choice about which model you'd rather defend.

Why OR can fail: it's all model

- OR's entire identification rests on $\widehat{\mu}_0(X)$ being correctly specified.
- Get the functional form wrong — a missing nonlinearity, a missing interaction — and OR is biased.
- Especially fragile in the post-period, where you're *extrapolating* from the control distribution of X to the treated distribution.
- If treated units have X-values rare among controls, OR is leaning on the model where the data are thin.

This is why IPW exists — as a different way to use X.

Lesson

2. Inverse Propen-sity Weighting

The propensity score, in one minute

- Definition: $p(X) = \Pr(D{=}1 \mid X)$ — the probability of treatment given covariates.
- Estimated by logit/probit **maximum likelihood**:

$$\widehat{\theta} = \arg\max_{\theta} \sum_{i} \left[\, D_i \log p(X_i;\theta) + (1 - D_i) \log(1 - p(X_i;\theta)) \,\right]$$

- **Rosenbaum-Rubin (1983)**: if D is unconfounded given X (a vector), it is unconfounded given $p(X)$ (a scalar).
  One number summarizes all of X.
- The pscore lives in $[0,1]$. Treated units have higher $p(X)$ on average. The pscore reorganizes the data along the dimension of selection.

The Rosenbaum-Rubin theorem (1983)

Statement

If treatment is unconfounded given X, i.e. $\{Y(0), Y(1)\} \perp\!\!\!\perp D \mid X$,
then it is also unconfounded given the propensity score $p(X)$ alone:

$$\{Y(0), Y(1)\} \perp\!\!\!\perp D \mid p(X).$$

- X is a vector. $p(X)$ is a scalar in $[0, 1]$. **One number compresses all the confounding information.**
- Operationally: matching, weighting, or balancing on $p(X)$ gives the same identification as on the full X.
- This is what licenses IPW. It does *not* license putting $p(X)$ on the right-hand side of a regression — that's a different use, with different consequences (Lesson 4 trap).

Abadie (2005): reweight the controls

- Idea: reweight control units by $\dfrac{p(X)}{1-p(X)}$ so the reweighted control sample *looks like* the treated sample in X.
- Then take a simple DiD on the reweighted sample.

The Abadie IPW-DiD estimator

$$\widehat{\delta}_{\text{IPW}} = \frac{1}{N_1} \sum_i \left(\frac{D_i - p(X_i)}{1 - p(X_i)} \right) \Delta Y_i$$

The operator $(D-p)/(1-p)$ gives weight $+1$ to treated units and weight $-p/(1-p)$ to controls — exactly the reweighting in line one. **Identification:** under conditional PT + common support, IPW recovers the ATT. Proof next slide.

Substitute $D{=}1$ and $D{=}0$: the operator does different things to each

The operator we care about:

$$\frac{D - p(X)}{1 - p(X)} \cdot \Delta Y$$

For a $D{=}1$ row:

$$\frac{1 - p(X)}{1 - p(X)} \cdot \Delta Y \; = \; \Delta Y$$

Just the treated unit's change. **No reweighting.** **For a $D{=}0$ row:**

$$\frac{0 - p(X)}{1 - p(X)} \cdot \Delta Y \; = \; -\frac{p(X)}{1 - p(X)} \cdot \Delta Y$$

Controls get reweighted by $p(X)/(1 - p(X))$ — and with a negative sign.

The reweighting only happens to the control group's first differences.
That's because it's the Y^0 that's missing, not the Y^1.

Take expectations: under conditional PT, the operator recovers the ATT

Treated piece ($D = 1$ rows contribute ΔY):

$$\mathbb{E}[\Delta Y \cdot \mathbb{1}\{D{=}1\}] \;=\; \Pr(D{=}1) \cdot \mathbb{E}[\, Y^1_{\text{post}} - Y^0_{\text{pre}} \mid D{=}1 \,]$$

Control piece ($D = 0$ rows contribute $-\frac{p(X)}{1-p(X)}\Delta Y$). By iterated expectations:

$$\begin{aligned} -\mathbb{E}\left[\tfrac{p(X)}{1-p(X)}\,\Delta Y \cdot \mathbb{1}\{D{=}0\}\right] \;&=\; -\mathbb{E}\left[\tfrac{p(X)}{1-p(X)} \cdot (1 - p(X)) \cdot \mathbb{E}[\Delta Y \mid D{=}0, X]\right] \\ &=\; -\mathbb{E}\big[p(X)\,\mathbb{E}[\Delta Y \mid D{=}0, X]\big] \end{aligned}$$

Apply conditional PT: $\mathbb{E}[\Delta Y^0 \mid D{=}0, X] = \mathbb{E}[\Delta Y^0 \mid D{=}1, X]$, so the control piece becomes

$$-\,\mathbb{E}\big[p(X)\,\mathbb{E}[Y^0_{\text{post}} - Y^0_{\text{pre}} \mid D{=}1, X]\big] \;=\; -\,\Pr(D{=}1) \cdot \mathbb{E}[\, Y^0_{\text{post}} - Y^0_{\text{pre}} \mid D{=}1 \,]$$

Add the two pieces: Y^0_{pre} cancels, leaving

$$\Pr(D{=}1) \cdot \mathbb{E}[\, Y^1_{\text{post}} - Y^0_{\text{post}} \mid D{=}1 \,] \;=\; \Pr(D{=}1) \cdot \text{ATT}$$

Normalize by $\Pr(D{=}1) \approx N_1/N$:

$$\widehat{\delta}_{\text{IPW}} \;=\; \frac{1}{N_1} \sum_i \frac{D_i - p(X_i)}{1 - p(X_i)}\,\Delta Y_i \;\to\; \text{ATT}. \quad \blacksquare$$

Common support is non-negotiable

- IPW's weights are $\dfrac{p(X)}{1-p(X)}$ for controls. As $p(X) \to 1$, the weight $\to \infty$.
- If some treated X-values have $p(X) \approx 1$, then NO control unit is comparable — and the weights are unstable.
- Standard responses:
  - Trim units with $p(X) > 0.95$ (or < 0.05).
  - Report sensitivity to the trim threshold.
  - In the limit, "weak overlap" breaks identification entirely. (Advanced deck.)

The pscore optimizer (IRLS, Newton-Raphson) can also misbehave near separation. Forward link to advanced.

Lesson

3. Doubly Robust DiD

Doubly robust: two chances to be right

- OR needs $\widehat{\mu}_0(X)$ correctly specified.
- IPW needs $\widehat{p}(X)$ correctly specified.
- Neither is foolproof. Each fails if its own model is wrong.

The Sant'Anna & Zhao (2020) move

Combine them. The **DR-DiD** estimator is consistent if *either* model is correct — not necessarily both.

Two chances to be right. That is what "doubly robust" means.

DR-DiD identification: three assumptions

Assumption 1 (Data)

Panel or repeated cross-sections with pre- and post-periods; treatment in post-period; covariates X pre-treatment.

Assumption 2 (Conditional parallel trends)

$$\mathbb{E}[\Delta Y(0) \mid D{=}1, X] \;=\; \mathbb{E}[\Delta Y(0) \mid D{=}0, X]$$

Assumption 3 (Common support)

$$0 \;<\; p(X) \;<\; 1 \text{ for almost every } X.$$

DR-DiD + these three $\Rightarrow$ ATT identified if either model is correct. Proof next slide.

The DR-DiD estimator

$$\widehat{\delta}_{\mathsf{DR}} = \mathbb{E}\left[\left(\frac{D}{\mathbb{E}[D]} - \frac{\frac{p(X)(1-D)}{1-p(X)}}{\mathbb{E}\left[\frac{p(X)(1-D)}{1-p(X)}\right]} \right) \left(\Delta Y - \mu_0(X)\right) \right]$$

- $p(X)$: propensity score model
- $\mu_0(X)$: outcome model fitted on *controls*: $\mu_0(X) = \mathbb{E}[\Delta Y \mid D{=}0, X]$
- The IPW weight reweights the comparison. The OR term centers the residual on the imputed counterfactual.
- If p is right, the IPW piece does the work. If μ_0 is right, the OR piece does the work. Proof next slide.

Proof: doubly robust — *either* model can be wrong

Let $w(X)$ denote the weight operator in $\widehat{\delta}_{\text{DR}} = \mathbb{E}[w(X)(\Delta Y - \mu_0(X))]$.

Case A: $p(X)$ **correct,** $\mu_0(X)$ **may be wrong.**

$$\widehat{\delta}_{\text{DR}} \;=\; \underbrace{\mathbb{E}[w(X)\,\Delta Y]}_{\text{IPW identifies ATT}} \;-\; \underbrace{\mathbb{E}[w(X)\,\mu_0(X)]}_{=\,0 \text{ in expectation}} \;=\; \text{ATT.}$$

The OR term vanishes because $\mathbb{E}[w(X) \mid X] = 0$ once $p(X)$ is correct. **IPW alone identifies the ATT.**

Case B: $\mu_0(X)$ **correct,** $p(X)$ **may be wrong.** Under conditional PT, the residual $\Delta Y - \mu_0(X)$ has conditional mean zero given $D{=}0, X$:

$$\mathbb{E}[\Delta Y - \mu_0(X) \mid D{=}0, X] \;=\; 0,$$

so the control piece zeroes regardless of weighting. The treated piece $\frac{D}{\mathbb{E}[D]}(\Delta Y - \mu_0(X))$ integrates directly to the ATT. **OR alone identifies the ATT.**

Either suffices — not both. That is "two chances to be right." ■

DR is *not* regression-with-pscore

- Look at the formula. $\widehat{p}(X)$ **appears as a weight, never as a regressor.**
- $\mu_0(X)$ models the outcome, but on *controls only* — to impute the counterfactual trend.
- The two pieces are combined by a re-centering and reweighting, NOT by adding $\widehat{p}$ to a regression's right-hand side.

The pscore is the tool. ***Using it as a weight is doubly robust.*** *Using it as a regressor is the trap from Lesson 4.*

Efficiency and the influence function

- Under conditional PT + common support, DR-DiD is **semiparametrically efficient**.
- No estimator of the ATT that uses only the same assumptions can have a smaller asymptotic variance.
- The asymptotic variance has a closed form via the *influence function* (this is what the `drdid` package computes for SEs).

Practical recipe

Pick X for confounding (Lesson 1). Estimate $\widehat{p}(X)$ by logit. Estimate $\widehat{\mu}_0(X)$ on controls. Plug into the DR-DiD formula. Influence-function SEs.

The modern frontier: double-debiased ML

Chernozhukov, Chetverikov, Demirer, Duflo, Hansen, Newey & Robins (2018), *The Econometrics Journal*.

Same DR architecture — model the outcome, model the treatment, combine them. Two extensions:

- **Replace logit / OLS with any ML learner.** Lasso, random forests, gradient boosting, neural nets. The DR formula stays.
- **Cross-fitting.** Estimate $\widehat{p}$ and $\widehat{\mu}_0$ on one half of the sample, plug in on the other. Swap. Keeps the influence-function argument valid even when the nuisance models overfit.

Same lesson, more flexible tools. DR is what makes the ML extension principled — without doubly-robustness, fitting nuisance models with ML would invalidate the standard errors.

Reading: Chernozhukov et al. (2018, Econometrics Journal). The DRDID + ML combination is the active frontier.

DR-DiD, in code

```
library(DRDID)

# Panel data: outcome y, treatment indicator d, time t, id, covariates X
result <- drdid(
  yname = "y",
  tname = "t",
  idname = "id",
  dname = "d",
  xformla = ~ x1 + x2 + x3, # covariates for BOTH p(X) and mu_0(X)
  data = panel_df,
  estMethod = "imp", # Sant'Anna & Zhao (2020) DR
  weightsname = NULL,
  boot = FALSE,
  inffunc = TRUE # influence-function SEs
)
summary(result)
```

Same X used for both models. Misspecify one — the other catches you.

A Monte Carlo to settle the rankings

- Compare four estimators on the same data: **TWFE** (with X as additive controls), **OR**, **IPW**, **DR**.
- **4 DGPs** (Sant'Anna's design): vary whether the OR model and the pscore model are correctly specified.
- **Sample size:** 1,000 units. **Replications:** 10,000.
- **Report:** bias, RMSE, SE, 95% coverage, CI length.

The question we want answered: when does DR pay off, and when does it break?

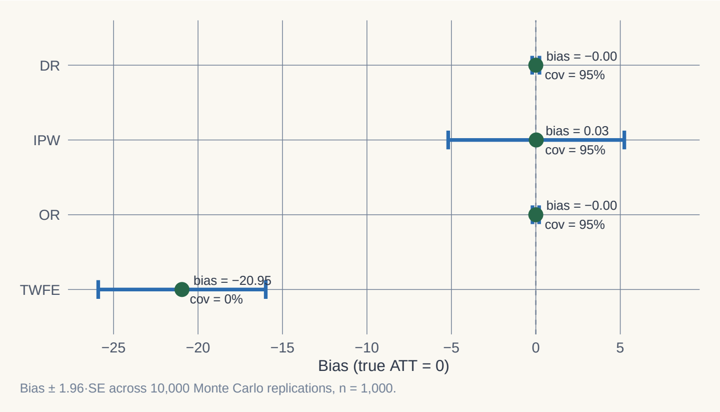
Driver code: `simulations/covariates/covariates_monte_carlo.{do,R}`

DGP 1 — both models correct

	Bias	RMSE	SE	Coverage	CI length
TWFE	-20.95	21.12	2.53	0.000	9.91
OR	-0.001	0.10	0.10	0.950	0.40
IPW	$+0.026$	2.77	2.66	0.952	10.44
DR	-0.001	0.11	0.11	0.947	0.41

*TWFE catastrophic (Bias ≈ -21; coverage zero). OR & DR both efficient (Bias ≈ 0, coverage 95%). IPW unbiased but **wide CIs**.*

DGP 1 — visualizing the sampling distribution



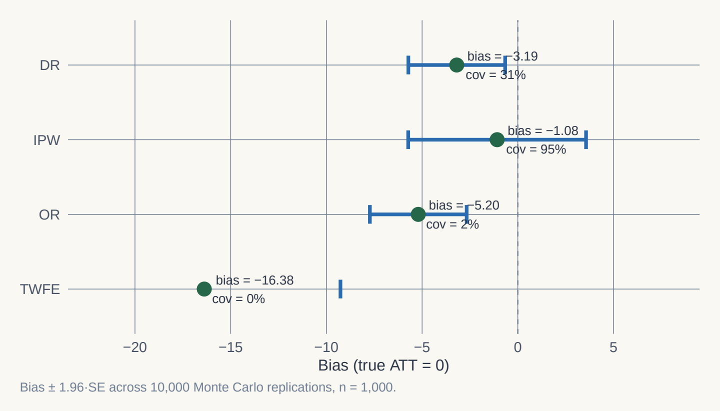
When both models are correct, OR and DR concentrate tightly on the truth. IPW is unbiased but noisy. TWFE is off the map.

DGP 4 — both models *wrong*

	Bias	RMSE	SE	Coverage	CI length
TWFE	-16.38	16.54	3.63	0.000	14.22
OR	-5.20	5.36	1.29	0.015	5.05
IPW	-1.08	2.66	2.37	0.949	9.31
DR	-3.19	3.45	1.29	0.308	5.07

DR is not magic. When both models fail, DR is biased and undercovers (0.31 vs. nominal 0.95). IPW alone now wins on coverage — it doesn't rely on the outcome model.

DGP 4 — the sampling distribution tells the same story



DR no longer hugs the truth. IPW is wider but centered. Doubly robust ≠ doubly unbeatable.

What the Monte Carlo teaches

- **TWFE with additive X:** never wins. Coverage is zero in every DGP. **Stop using it as a default.**
- **OR:** efficient when the outcome model is right. Disaster when it's wrong (Bias -5, coverage 1.5%).
- **IPW:** less efficient but *robust to outcome misspecification*. The unsexy workhorse.
- **DR:** best of both *when at least one is right*. Not better than IPW when *both* are wrong.

"Doubly robust" means robust to one misspecification. Not two. Not magic.

Replication: `simulations/covariates/covariates_monte_carlo.{do,R}`; figures `mc_dr_1.png`, `mc_dr_2.png`.

Closing addendum: DR is *not* regression-plus-pscore

A common misdescription you'll hear:

> *Doubly robust DiD is just outcome regression plus the propensity score as another control.*

— Heard at more than one seminar

No. That description conflates three different things:

- what **outcome regression** actually does (model $\mu_0(X)$ on controls; impute counterfactuals),
- what **IPW** actually does (use $\widehat{p}(X)$ as a *weight*, not a regressor),
- what **DR** actually does (combine the two).

Adding $\widehat{p}(X)$ as a regressor on the right-hand side is its own — different — estimator. It's a trap. The next few slides walk through exactly why.

A natural temptation

> *If the propensity score is the one-scalar summary of confounding, why not just add it as a regressor? Done.*

— The audience, right now

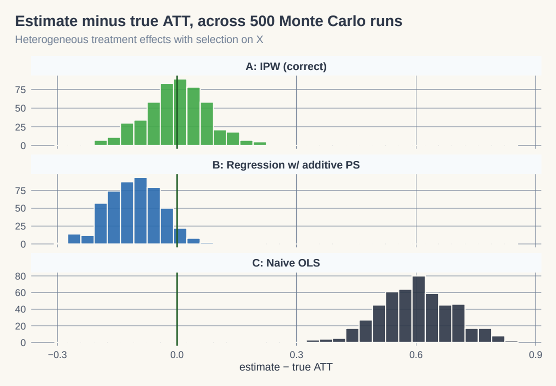
- It sounds right. Rosenbaum-Rubin says *one scalar suffices*. Regression handles scalars.
- So: $Y_i = \alpha + \tau D_i + \gamma \widehat{p}(X_i) + \varepsilon_i$
- Surely this recovers τ as the ATT? **No.** And the reason matters.

What that regression actually assumes

$$Y_i = \alpha + \tau D_i + \gamma\,\widehat{p}(X_i) + \varepsilon_i$$

- No $D \times X$ interaction. No $D \times \widehat{p}$ interaction. **The treatment effect is forced constant** across the entire range of $\widehat{p}(X)$.
- Equivalently: the same dose of policy produces the same effect on a person with $\widehat{p}(X) = 0.10$ as on a person with $\widehat{p}(X) = 0.90$.
- That's an assumption about **the outcome side** — one that Rosenbaum-Rubin says nothing about.
- Rosenbaum-Rubin gives you a valid *reweighting* tool. It does not authorize regression-on-pscore.

A simulation makes it visible



500 reps. Heterogeneous $\tau(X) = 1 + 0.5\,X_1$. Method A (IPW reweighting): bias ≈ 0. Method B (regression with $\widehat{p}$): ***bias*** *$\approx -10\%$.*

Code: `glasgow/decks/code/IPW-Demo/ipw_simulation.R`.

The simulation, one screen of R

```r
N <- 1000
X1 <- rnorm(N); X2 <- rnorm(N)
p <- plogis(0.5*X1 + 0.3*X2 - 0.2) # true propensity
D <- rbinom(N, 1, p)
Y0 <- X1 + 0.5*X2 + rnorm(N)
Y1 <- Y0 + (1 + 0.5*X1) # heterogeneous tau(X)
Y <- D*Y1 + (1-D)*Y0

phat <- glm(D ~ X1 + X2, family=binomial)$fitted.values

# Method A: IPW -- reweight controls
w <- ifelse(D==1, 1, phat/(1-phat))
A <- mean(Y[D==1]) - weighted.mean(Y[D==0], w[D==0])

# Method B: regression with pscore as a control
B <- coef(lm(Y ~ D + phat))["D"] # <-- the trap
```

Full simulation: `decks/code/IPW-Demo/ipw_simulation.R`

Weight versus regressor — different uses of the same scalar

- The propensity score is a **tool**. The same scalar can be used as:
  - a **weight** (Method A: reweight controls) — recovers ATT
  - a **regressor** (Method B: add as control) — imposes homogeneous effects
- Same number. Different uses. *Different identifying assumptions.*
- Method B's bias is not noise. It is the heterogeneity in $\tau(X)$ that the regression cannot see.

The lesson before DR

"Doubly robust" is sometimes mis-summarized as "IPW plus a regression control."

That's wrong. DR is something else — it uses the pscore as a weight, AND it uses an outcome model.
But it does NOT use the pscore as a regressor.

Name the disease: *additive-control-side bias*

Whenever you put a covariate on the right-hand side of an outcome regression *additively* (no $D \times X$ interaction), you force:

τ is constant across all values of that covariate.

- Method B (Lesson 4): the covariate is $\widehat{p}(X)$ — a scalar summary of X.
  Force constant τ across $\widehat{p}(X)$. Heterogeneity in $\tau(X)$ is invisible.
- TWFE-with-controls (Lesson 2): the covariate is X itself.
  Force constant τ across X. Heterogeneity in $\tau(X)$ is invisible.
- **Same disease, different costume.** Caetano & Callaway (2024) name the TWFE version *hidden linearity bias*. It's the same root.

DR's escape: use X and $\widehat{p}(X)$ as weights and to model the counterfactual $\mu_0(X)$ on controls only — never as additive regressors of Y.

LESSON

4. Hidden Linearity Bias

Caetano & Callaway 2024

TWFE-with-controls has two distinct problems

The workhorse specification $Y_{it} = \alpha_i + \gamma_t + \tau D_{it} + X_{it}'\beta + \varepsilon_{it}$ is biased for two *separate* reasons:

1. **The FE transformation throws away what you wanted to control for.**
   First-differencing X_{it} leaves only ΔX. The *level* of X is gone. Time-invariant Z is gone entirely. (Caetano-Callaway 2024 — this lesson.)
2. **What's left enters additively, forcing constant τ.**
   Whatever covariate survives the FE transformation sits on the RHS as an additive regressor with no $D \times X$ interaction. Same disease as Method B (Lesson 4).

This lesson covers problem (1). Problem (2) is the L4 callback we'll close on.

An applied problem you've already run

> *TWFE with time-varying controls is the workhorse DiD specification. Almost every DiD paper runs it.*

— Anybody who's seen a job-market paper

- Standard recipe: include population, income, demographic shares, region indicators, etc. on the right-hand side. Run TWFE. Done.
- **What if I told you that regression doesn't control for what you think it controls for?**
- Caetano & Callaway (2024, R&R *REStat*): yes. The fixed-effect transformation quietly drops your X levels and your time-invariant Z.

An application: stand-your-ground laws (Cheng & Hoekstra 2013)

- 50 states, 2000–2010. 20 states adopt stand-your-ground laws. Outcome: log homicide rate.
- Treated states differ from controls in **levels**:
  - South dummy: std. diff. $= 0.81$ (large)
  - Poverty rate: std. diff. $= 1.06$ (huge)
  - Median income: std. diff. $= -1.08$ (huge)
- But the differences in **changes** of these covariates are small.

The headline result

TWFE with covariates: $\widehat{\alpha} = +0.067$, significant.
AIPW with the right covariates: $\widehat{\text{ATT}} \approx 0$, insignificant.

The positive TWFE result is largely artefact of hidden linearity bias.

What TWFE actually controls for, after the FE transformation

You wrote down this model:

$$Y_{it} = \alpha_i + \gamma_t + \tau D_{it} + X_{it}'\beta + Z_i'\delta + \varepsilon_{it}$$

First-difference (or within-transform) to kill α_i:

$$\Delta Y_{it} = \tilde{\gamma}_t + \tau\,\Delta D_{it} + \Delta X_{it}'\beta + \Delta\varepsilon_{it}$$

- α_i vanishes (good).
- Z_i vanishes too (time-invariant). **Region: gone.**
- X_{it} becomes ΔX_{it}. **Population level: gone. Only population growth remains.**
- Your "controls" are now *changes* in X, not levels.

What you should be controlling for

The correct first-differenced model for $\Delta Y(0)$ (Caetano-Callaway Eq. 4):

$$\Delta Y_{it}(0) = \widetilde{\gamma}_t + \underbrace{Z_i'\widetilde{\delta}_t}_{\text{time-inv}} + \underbrace{\Delta X_{it}'\beta_t}_{\text{changes}} + \underbrace{X_{i,t-1}'\widetilde{\beta}_t}_{\text{level}} + \Delta\varepsilon_{it}$$

- **Time-invariant** Z (region) belongs on the right — because trends differ by region.
- **Pre-period level** $X_{i,t-1}$ belongs on the right — because the level of population predicts the trend in homicides.
- TWFE drops both. **Hidden linearity bias.**

The L4 callback: problem (2) is the additive-RHS disease

We've covered problem (1) above — the FE transformation throws away levels.
Problem (2) is the L4 disease, present in TWFE-with-controls too.

Estimator	**What goes on RHS additively**	**What gets forced constant**
Method B (L4)	$\widehat{p}(X)$ as control	τ across $\widehat{p}(X)$
TWFE-with-controls (L2, L6)	ΔX as control	τ across ΔX
General formulation	$W = f(X)$ as additive RHS regressor	τ across W

- Problem (1) is C&C's contribution: *what* you control for is wrong.
- Problem (2) is the L4 disease: *how* you control for it is wrong.
- TWFE-with-controls catches both. DR escapes both — using X and $\widehat{p}(X)$ as **weights**, not as additive regressors.

The fix: AIPW with $(X_{t-1},\, Z)$

Caetano-Callaway's recommended estimator:

$$\widehat{\text{ATT}} = \widehat{\mathbb{E}}_{D=1}[\Delta Y - \widehat{m}_0(X_{t-1}, Z)]$$

- $\widehat{m}_0(X_{t-1}, Z)$: outcome regression of ΔY on *pre-treatment level* of X and *time-invariant* Z — fit on **controls only**.
- **Why X_{t-1}, not X_t?** Because X_t in the post-period could itself be affected by treatment. The pre-treatment level is uncontaminated.
- Combine with IPW on a generalized propensity score (same conditioning set) for full doubly-robust safety.
- CC's fully preferred spec also includes $\Delta X = X_t - X_{t-1}$ for dimension reduction. R: `pte`, `twfeweights`.

Pre-treatment X is uncontaminated — post-treatment X is not

> *Why X_{t-1} and not X_t? You're throwing away the most recent data point.*

— A student, raising a hand

- X_t sits **downstream of treatment**. If the policy moves X — even a little — you've controlled for a *post-treatment outcome*.
- Conditioning on a post-treatment variable doesn't reduce bias. It **re-introduces** bias — the bad-controls problem (Angrist–Pischke 2009, Ch. 3).
- X_{t-1} is settled before D is assigned. It carries the selection information without carrying any of the treatment.

The rule generalizes: condition on the latest pre-treatment value, never the contemporaneous one.

Caetano & Callaway (2024), §3; classic "bad controls" point goes back to Rosenbaum (1984).

Stand-your-ground revisited — the numbers

Specification	$\widehat{\text{ATT}}$	SE
TWFE with ΔX (the standard recipe)	$+0.067$	0.026
Reg. adj. with $\Delta X + Z$	$+0.106$	0.058
Reg. adj. with $X_{t-1} + Z$ (*Caetano-Callaway preferred*)	$+0.019$	0.043
Reg. adj. with $\Delta X + X_{t-1} + Z$	$+0.078$	0.180

- The TWFE positive effect ($+6.7\%$) shrinks to near-zero ($+1.9\%$) when you control for *levels* of X and Z.
- The bias was hidden in plain sight — the regression compiled fine. It just wasn't controlling for what we thought.

Lesson

5. Compositional Change

Most applied work doesn't have a panel

> *Panels are expensive. Repeated cross-sections are how a lot of applied work actually gets done.*

— Anyone who's ever priced a long household-survey panel

- **US workhorses:** CPS, NLSY, ACS, CEX — repeated cross-sections, not panels.
- **Development workhorses:** LSMS (World Bank), DHS (USAID), MICS (UNICEF), IFLS (Indonesia), IHDS (India) — repeated rounds, mostly different households each wave.
- Same DiD recipe runs on all of them. **Same compositional-change problem too.**

Hong (2013) found it in the CEX. Sant'Anna & Xu's own application is shipments through the South Africa–Mozambique trade corridor. The disease travels.

Two flavors of "X moves"

Flavor	What's moving	Estimator
Lesson 6	Same units, X_{it} moves over time	Caetano-Callaway AIPW
Lesson 7	New units each period, X distribution moves	Hong / Sant'Anna-Xu

- Both break unconditional PT *in finite samples* even when it holds in the population.
- L6 (panel): the FE transformation loses the X level. L7 (repeated cross-sections): the sample composition shifts.
- Different fixes. Same family of concerns: X **is moving, and your estimator pretends it isn't.**

Hong (2013): Napster, internet users, and music expenditure

- Repeated cross-sections from the CEX, 1996–2001.
- In 1996–97, internet users were a small, atypical slice — younger, higher-income, more urban, more educated.
- By 2000–01, internet users looked much more like the general population.
- Compare “internet users” to “non-users,” 1996 vs. 2001.
  Did Napster reduce music spending — or did the composition of internet users change?
- Hong’s answer: the naive DiD confounds the two. The compositional shift accounts for much of the apparent “Napster effect.”

Hong (2013), Journal of Applied Econometrics 28(2): 297–324.

Hong's Table I — the internet-user composition shifts

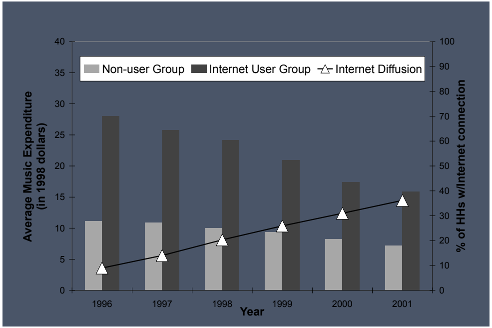
Table 77: Changes between Internet and non-Internet users over time

Year	1997		1998		1999		2000	
	Internet user	Non-user	Internet user	Non-user	Internet user	Non-user	Internet user	Non-user
Demographics								
Age	40.2	49.0	42.3	49.0	44.1	49.4	44.3	49.9
Income	$52,887	$30,459	$51,995	$26,189	$49,970	$26,649	$47,510	$26,336
High school graduate	0.18	0.31	0.17	0.32	0.21	0.32	0.22	0.33
Some college	0.37	0.28	0.35	0.27	0.34	0.27	0.36	0.27
College grad	0.43	0.21	0.45	0.21	0.42	0.20	0.37	0.20
Manager	0.16	0.08	0.16	0.08	0.14	0.08	0.14	0.07

Sample means from the Consumer Expenditure Survey from Hong [2013].

1997 → 2000: mean age 40 → 44, income $53k → $48k, college share 43% → 37%. That gap is what naive DiD attributes to "Napster."

Hong’s Figures 1 & 3 — diffusion and absorption of zero-spenders



Left: internet diffusion (top axis) and average quarterly music expenditure (bottom).
Right: stacked share of households by Internet adoption × music-spending status — the user pool absorbs more zero-spenders as it grows.

So why do we need Sant'Anna & Xu given Hong showed it?

Hong (2013) gave us the intuition — and a single empirical example. To use this in your own work you need three things Hong didn't formalize:

1. **A target estimand.** What *is* “the” ATT under compositional shift? Is it the average over the post-period sample (your data) or the pre-period sample? These are different parameters.
2. **A test.** Hong showed his data had the problem. How do *you* know yours does, without the benefit of a second empirical paper?
3. **A correction.** If your test rejects, what estimator should you report? Hong didn't give you one — he flagged the problem.

The story stays **Hong’s.**
The toolkit is **new.**

Sant'Anna & Xu name the target: the ATT for the post-period treated

> *What is "the" ATT when the sample composition is moving?*

— Anyone trying to use Hong in their own paper

The SX target

$\tau \; = \; \mathbb{E}\big[\, \tau(X) \,\big|\, D{=}1,\, T{=}1 \,\big]$ — ATT averaged over the **post-period treated** X-distribution.

- **Why this target?** It's the population that actually got treated — the policy-relevant parameter.
- Naive DiD targets a *moving* mixture — whoever happened to be in each period's sample.
- SX pin the target to a single population so you know what number you're trying to estimate.

The post-period treated is the population the policy is for

> *Why average $\tau(X)$ over the post-period treated? Why not the pre-period treated, or both?*

— A student, raising a hand

- The post-period treated is **who actually got the policy**. Whatever number you report is implicitly about them.
- Sant'Anna–Zhao's τ_{sz} averages over the *pooled* (pre $+$ post) treated — a population that doesn't correspond to any policy decision.
- If $\tau(X)$ is heterogeneous and the post-period treated are a different mix from the pre-period treated, τ_{sz} answers a question **nobody asked**.

Pick the estimand that matches the counterfactual you actually want to compare.

Sant'Anna & Xu (2023), §2; bias decomposition Eq. 2.4.

SX's Hausman test: compare two doubly-robust estimators

The test

Compute the standard DR-DiD estimator $\widehat{\tau}_{\text{dr}}$. Compute the SX composition-corrected DR estimator $\widehat{\tau}_{\text{sx}}$. Compare.

- Under the null (no compositional shift), the two DR estimators agree — both consistent for the same τ.
- Under the alternative, they diverge — and the gap quantifies how much your point estimate is doing compositional vs. behavioral work.
- Standard χ^2 inference on $\widehat{\tau}_{\text{dr}} - \widehat{\tau}_{\text{sx}}$.

This is the move Hong didn't have. He showed his data had the problem. The Hausman test lets you check yours.

The SX estimator: reweight to the post-period treated, then DR-DiD

Mechanics

For each period t, reweight observations so the within-period X-distribution matches the **post-period treated** distribution. Then run DR-DiD on the reweighted panel.

- Weights come from a 4-cell generalized propensity score: $\Pr(D, T \mid X)$, not the simple f_{ref}/f_t density ratio.
- **Doubly robust** — consistent if either the PS or the outcome model is right.
- Built on the IPW machinery from Lesson 3 $+$ the DR machinery from Lesson 5.

SX in the wild: tariff reductions and *bribery* at the SADC border

- **Setting:** Sequeira (2016) studied SADC tariff reductions (2008, $\sim$5% cut on selected products) at the South Africa–Mozambique border. Some products were treated; others weren't.
- **Outcome:** *bribery* at customs (probability, amount, amount/value, amount/tonne).
- **Data:** repeated cross-sections of shipment records, 2007–2013. $\sim$80% of pre-treatment shipments involved bribery.
- **SX's question:** does compositional change in *which shipments get observed* contaminate Sequeira's DiD?

Hausman test result

$p = 0.338$ **— fail to reject.** Composition is not biting here.

DR-DiD and SX composition-corrected DR agree. Sequeira's finding holds: tariff reductions reduce bribery.

Failing to reject *is* the validation — that's the point

> *The Hausman test failed to reject. So SX wasn't really doing anything here, right?*

— A student, raising a hand

- Wrong frame. The test isn't a switch that turns SX "on" when it rejects. **It's a diagnostic that tells you whether composition mattered for your ATT.**
- In Sequeira: $p = 0.338$ *is* the result. It says "the composition correction would have moved the number, but didn't move it enough to matter."
- In Hong's Napster setting — richer adopters in 1996, mass-market by 2001 — it would have rejected. **Same test. Different application. Different verdict.**

You don't get to know which world you're in until you run the test. That's why you run it.

Sant'Anna & Xu (2023), Theorem 4.1; Sequeira (2016) for the application.

Use the SX test to *validate*, not to *select*

- **Wrong:** run Hausman as a pre-test; pick $\widehat{\tau}_{\text{sx}}$ if it rejects, $\widehat{\tau}_{\text{dr}}$ if not.
- **Why wrong:** pre-test-then-select distorts inference (Roth 2022; Guggenberger 2010). Standard errors lose coverage.
- **Right:** if you're worried about composition, *just report* $\widehat{\tau}_{sx}$. Use the test to communicate whether composition was a concern, not to choose which estimate to print.

The robust estimator costs efficiency. Worth paying when composition might shift.

When compositional change matters — and it's almost always

- **Long-horizon US survey data.** CPS, NLSY, ACS, CEX — sample composition drifts as response rates, demographics, frame definitions change.
- **Developing-country household surveys.** LSMS, DHS, MICS, IFLS, IHDS — rotating cross-sections, expanding sampling frames between rounds.
- **Administrative outcomes with selective coverage.** Who enters the data may itself respond to policy (Medicaid take-up, charter-school enrollment, customs declarations).
- **Policy-diffusion settings.** New technology, new programs, new tariff regimes — the adopter pool shifts.
- **Long pre-periods in event studies.** The further back you go, the more likely the comparison population has shifted.

First defense: check whether X moved across time. Then run the Hausman test. Then report both estimates.

LESSON

6. Choosing & checking X

The practical close — picking, balance, and when controls are unnecessary

Picking X: theory or LASSO on $\Delta Y(0)$

You want X that (i) predicts the treatment *and* (ii) predicts the untreated trend.
Pre-treatment only — post-treatment X can be affected by D and will leak.

Two paths to choose which X:

- **Theory / common sense.** Variables you believe drive both selection and the untreated trend. This is how most of it actually gets done.
- **Data-driven.** LASSO (or another selector) on

  $$\Delta Y(0)_i \ \sim \ X_i, \quad \text{fit on untreated unit-periods only.}$$

  Keep what LASSO keeps.

The regressand is $\boldsymbol{\Delta Y(0)}$. Not $Y(1)$. Not post-treatment Y — that's $Y(1)$, and using it lets D leak through the back door. This is **Heckman-Ichimura-Todd (1997)** logic: the data can only teach you which X's predict the untreated counterfactual trend.

Beyond that — functional form, interactions, transformations — the data is mostly silent.

If X is balanced, you don't need to control for it

Same logic as an RCT: balanced X across treatment and control makes the comparison fair without conditioning.

Worked example. Let sex be the only confounder. Within sex, the untreated trends are:

$$\Delta Y^0_{\text{men}} \;=\; 10-6 \;=\; 4 \qquad\qquad \Delta Y^0_{\text{women}} \;=\; 8-5 \;=\; 3.$$

Balanced — both groups 50% male.

$$\mathbb{E}[\Delta Y^0 \mid D{=}1] \;=\; 0.5(4)+0.5(3) \;=\; 3.5 \;=\; \mathbb{E}[\Delta Y^0 \mid D{=}0].$$

Unconditional PT holds. You didn't condition on sex — and you didn't need to.

Imbalance turns a confounder into bias

Same untreated trends as before — men $+4$, women $+3$ — but now the groups have different sex composition:

Imbalanced — treated 75% male, control 25% male.

$$\begin{aligned}\mathbb{E}[\Delta Y^0 \mid D{=}1] &= 0.75(4) + 0.25(3) = 3.75\\ \mathbb{E}[\Delta Y^0 \mid D{=}0] &= 0.25(4) + 0.75(3) = 3.25.\end{aligned}$$

Gap $=$ 0.5. That's the bias a naive DiD on ΔY would carry.

Takeaway. Balance is what licenses unconditional PT. Imbalance turns a Y^0-predictor into a bias on ΔY^0.

Check the balance: normalized differences

Before you trust your X, check that the treated and control groups are *actually* comparable on it.

Normalized difference

$$\text{Norm. Diff}_X = \frac{\overline{X}_T - \overline{X}_C}{\sqrt{(S_T^2 + S_C^2)/2}}$$

- A unit-free imbalance measure — comparable across variables.
- **Rule of thumb: < 0.25.** Above that, classical OLS/IPW have trouble.
- Unlike a t-statistic, it doesn't shrink to zero just because the sample is small.

Imbens & Rubin (2015), Causal Inference for Statistics, Social, and Biomedical Sciences, Ch. 14.

What balance tells you (and what it doesn't)

A confounder has *two* jobs:

1. $X \to D$: different distribution in treated vs. control.
2. $X \to \Delta\mathbb{E}[Y^0]$: predicts the untreated outcome trend.

The balance check only diagnoses (1).

- Pick X theoretically — variables you believe predict Y^0 trends.
- *Then* run normalized differences to see whether they're differentially distributed.
- Imbalance + predictiveness $\Rightarrow$ controlling matters. Imbalance without predictiveness $\Rightarrow$ noise.

Don't over-spend on already-balanced X

Adding controls in OLS DiD is cheap. In IPW or DR, it isn't.

The propensity score is fit by logistic regression, which is fragile in high dimensions:

Events-per-variable rule

Roughly **5–10 events per covariate** for stable logistic estimates, where "events" $=$ treated units.

Implication. A balanced X that doesn't strongly predict ΔY^0 doesn't deserve a slot in the pscore model. The balance table tells you where to spend the budget. *EPV guideline: Peduzzi et al. (1996), J. Clin. Epidemiol.*

A balance table in the wild

Table 4: Covariate Balance Statistics

	Unweighted			Weighted		
Variable	Non-Adopt	Adopt	Norm. Diff.	Non-Adopt	Adopt	Norm. Diff.
2013 Covariate Levels						
% Female	27.94	28.32	0.19	29.89	30.13	0.14
% White	47.43	52.50	0.62	46.07	47.76	0.21
% Hispanic	5.86	4.75	-0.14	10.06	11.35	0.13
Unemployment Rate	7.10	7.77	0.25	6.98	8.00	0.50
Poverty Rate	18.54	16.22	-0.35	17.22	15.29	-0.37
Median Income	43.38	48.00	0.41	49.28	57.83	0.68
2014 - 2013 Covariate Differences						
% Female	-0.11	-0.13	-0.06	-0.05	-0.04	0.10
% White	-0.29	-0.35	-0.14	-0.29	-0.27	0.07
% Hispanic	0.11	0.11	-0.01	0.14	0.20	0.33
Unemployment Rate	-1.09	-1.26	-0.24	-1.08	-1.36	-0.54
Poverty Rate	-0.51	-0.28	0.13	-0.41	-0.35	0.04
Median Income	1.13	1.04	-0.04	1.11	1.73	0.32

This table reports the covariate balance between adopting and non-adopting states. In the top panel, we report the averages and standardized differences of each variable, measured in 2013, by adoption status. In the bottom panel we report the average and standardized differences of the county-level long differences between 2014 and 2013 of each variable. We report both weighted and unweighted measures of the averages to correspond to the different estimation methods of including covariates in a 2×2 setting.

Every variable, its treated mean, its control mean, the normalized difference. One column tells you whether the design will make you sweat.

The empirical crisis of the 1970s

Empirical economics in the 1970s did not have a strong reputation.

- Most empirical work was a postscript to a theory paper. Sargent on rational expectations is a representative case — the regression was an afterthought.
- IV in the 1970s could be done without even *reporting* the instrument. *Orley Ashenfelter and Janet Currie, in conversation.*
- Outlandish estimates were routine and rarely interrogated.

There was a sense that econometric methods couldn't quite be trusted — and no shared standard for adjudicating that.

The Lewis smell test

H. Gregg Lewis (Chicago) studied union wage differentials. In his book reviewing the literature, he simply *ignored* estimates he couldn't justify — particularly when they came from IV or Heckman selection corrections.

He preferred OLS with controls and a story you could defend.

Card called this the "Lewis smell test."

If you could get a result past Lewis's skepticism, that was worth noting. Otherwise — skip it.

The other world: program evaluation

There was one corner of empirical work that took the question seriously.

- Federal job-training programs in the 1970s — evaluations commissioned by the Department of Labor.
- Princeton's **Industrial Relations Section** — older than the Econ department — ran the empirical agenda.
- Job training was a *totem* in the Section. Heckman used it as a recurring metaphor (including the 1997 ReStud with Ichimura, Todd, and the later Smith papers).

Two worlds: empirics-as-afterthought, and program evaluation. The credibility gap was huge.

Orley and Card's students

Orley was at the center of Princeton's IRS. David Card — newly arrived as junior faculty in the early 1980s — slid onto LaLonde's and Angrist's dissertation committees. He gave students problems.

- **Ron Oaxaca** $\rightarrow$ the wage decomposition (1973). *Orley.*
- **Bob LaLonde** $\rightarrow$ the job-training paper (1986). *Orley and Card.*
- **Josh Angrist** $\rightarrow$ Vietnam draft-lottery IV (1990). *Orley and Card.*

Angrist once said: "I wish I had written LaLonde."

Foundational papers from one office.

The NSW experiment, and Orley's suggestion

The National Supported Work Demonstration (NSW). Late-1970s federal job-training program run by MDRC. Hard-to-employ adults. **Randomly assigned** treatment and control.

Orley to LaLonde:

> *"Estimate the ATT from the experiment — that's the truth.*
> *Then drop the experimental control. Replace it with survey data.*
> *Re-run the contemporary econometric methods. See what you get."*

A placebo-style design before placebo designs were a thing. A direct honesty test.

What LaLonde found (1986)

The truth from the RCT: ATT $\approx +\$800$.

What observational methods gave: all over the map. Wrong magnitudes. Often *wrong signs*.

The conclusion — and it cratered the credibility of observational program evaluation:

The standard econometric methods of the day could not be trusted to recover known causal effects.

Dehejia & Wahba (1999, 2002): the rescue

Mid-1990s. Harvard. **Imbens and Rubin's causal-inference class.** Students include Rajeev Dehejia, Sadek Wahba, Bruce Sacerdote, (later) David Autor.

- Rosenbaum–Rubin (1983) introduced the propensity score. *LaLonde didn't use it.*
- Dehejia & Wahba revisit the same data with **propensity-score matching**. They restrict to a sample with *two years* of pre-treatment earnings.
- On that sample they recover $\$1{,}794$ — very close to the experimental ATT.

Smith & Todd (2005) later debated the sample-selection choice. The DW result held up enough for the literature to take "conditional PT" seriously.

A closing example: what does conditional PT actually buy you?

We've spent the afternoon *assuming* conditional PT is more credible than unconditional PT.

- Is it?
- Is there *empirical* evidence that conditioning on X recovers a known truth?

Yes. The canonical case is LaLonde. We'll now run four observational specifications on the non-experimental panel and see which ones recover $\$1,794$.

The LaLonde panel

Data structure (Dehejia & Wahba's two-year-pre version):

- Random assignment to NSW job training. Outcome: real 1978 earnings.
- Years on record: `74`, `75`, `78`.

The experimental panel gives the truth. The non-experimental panel (NSW treated + CPS controls) is our test case.

The benchmark: $1,794

$$\widehat{\text{ATT}}^{\text{RCT}} = \overline{Y}^{T}_{78} - \overline{Y}^{C}_{78} = 6{,}349 - 4{,}555 = \$1{,}794$$

Post-period difference in means on the experimental sample. Random assignment makes this the truth.

The challenge: swap the experimental controls for CPS

Build a non-experimental panel.

- Keep the NSW *treated* group.
- Replace the experimental controls with a random sample of CPS households.
- Selection into NSW is sharply negative — trainees had lower pre-earnings than the CPS pool.

Can observational DiD recover $1,794 on this panel?

Spec 0: naive DiD, no covariates

$$Y_{it} \;=\; \alpha \;+\; \beta_1\,\text{treat}_i \;+\; \beta_2\,\text{post}_t \;+\; \tau\,(\text{treat}_i \times \text{post}_t) \;+\; \varepsilon_{it}$$

No X anywhere. $\widehat{\tau}$ is the DiD coefficient on the interaction.

Spec 0 in Stata

```
use lalonde_nonexp_panel.dta, clear
gen post = (year == 78)
gen treat = (ever_treated == 1)

reg re i.treat##i.post
```

Spec 0 result: $3,522 — way off

Estimate	
RCT benchmark	$\$1,794$
Spec 0 (no covariates)	$\$3,522$
Gap	$+\$1,728$

Doubles the truth. The non-experimental selection is doing real damage.

Spec A: add the Xs as additive controls

$$Y_{it} \;=\; \alpha + \beta_1\,\text{treat}_i + \beta_2\,\text{post}_t + \tau\,(\text{treat}_i \times \text{post}_t) \;+\; X_i'\gamma \;+\; \varepsilon_{it}$$

Twelve covariates, the standard LaLonde set:

`age, agesq, educ, educsq, marr, nodegree, black, hisp, re74, re75, u74, u75`

The natural thing to try. Surely conditioning on age, education, race, and pre-earnings fixes it?

Spec A in Stata

```
global X "age agesq educ educsq marr nodegree black hisp re74 re75 u74 u75"

reg re i.treat##i.post $X
```

Spec A in R

```r
library(fixest)

feols(re ~ treat*post + age + agesq + educ + educsq
                   + marr + nodegree + black + hisp
                   + re74 + re75 + u74 + u75,
      data = d)
```

Spec A in Python

```
import statsmodels.formula.api as smf

X = ("age + agesq + educ + educsq + marr + nodegree "
     "+ black + hisp + re74 + re75 + u74 + u75")

mod = smf.ols(f"re ~ treat*post + {X}", data=d).fit()
print(mod.params["treat:post"])
```

Spec A result: \$3,522 — *identical* to Spec 0?!

RCT benchmark	$\$1{,}794$
Spec 0 (no covariates)	$\$3{,}522$
Spec A (additive X)	$\$3{,}522$

Why? Most of these Xs are *time-invariant* within a person — age (within a window), education, race, pre-earnings.

Additive X on a panel with time-invariant covariates does nothing for the DiD coefficient. You need to let X's effect *move* in the post period.

Spec B: let X's effect interact with post

$$Y_{it} = \alpha + \beta_1\,\text{treat}_i + \beta_2\,\text{post}_t + \tau\,(\text{treat}_i \times \text{post}_t) \;+\; X_i'\gamma_1 \;+\; (\text{post}_t \cdot X_i)'\gamma_2 \;+\; \varepsilon_{it}$$

- The γ_1 vector: X's baseline effect (pre period).
- The γ_2 vector: how X's effect *shifts* in post.

This is what saturating X with the time index actually means. The post-period gets its own X-coefficient.

Spec B in Stata

```
global X "age agesq educ educsq marr nodegree black hisp re74 re75 u74 u75"

reg re i.treat##i.post i.post##(c.($X))
```

Spec B in R

```r
library(fixest)

feols(re ~ treat*post
        + post*(age + agesq + educ + educsq
                + marr + nodegree + black + hisp
                + re74 + re75 + u74 + u75),
      data = d)
```

Spec B in Python

```
import statsmodels.formula.api as smf

X = ("age + agesq + educ + educsq + marr + nodegree "
     "+ black + hisp + re74 + re75 + u74 + u75")

mod = smf.ols(f"re ~ treat*post + post*({X})", data=d).fit()
print(mod.params["treat:post"])
```

Spec B result: $1,184 — much closer to $1,794

RCT benchmark	$\$1{,}794$
Spec 0 (no covariates)	$\$3{,}522$
Spec A (additive X)	$\$3{,}522$
Spec B (post$\times X$)	$\$1{,}184$

Not exact. But the movement is toward the truth, not away from it. From $1,728 above to $610 below.

Spec C: heterogeneous τ in X — the saturated approach

The motivation. If the treatment effect itself depends on X (e.g., training works better for younger workers), then *no* single $\hat{\tau}$ coefficient is the ATT. You need to model the heterogeneity and average correctly.

The two-step regression-adjustment / OR estimator (Heckman, Ichimura & Todd, 1997):

1. Fit $\Delta Y = \alpha + X'\gamma + \nu$ on *controls only*. (Models how $Y(0)$ trends as a function of X.)
2. Predict the counterfactual $\widehat{\Delta Y_i}$ for each treated unit. Compute $\hat{\tau}_i = \Delta Y_i - \widehat{\Delta Y_i}$.
3. $\widehat{\text{ATT}} = \frac{1}{N_{tr}} \sum_{i:D_i=1} \hat{\tau}_i$.

Equivalent (but slower): run a fully saturated regression $Y \sim$ treat·post·X, predict counterfactual, average over the treated X distribution.

Spec C in Stata

```
* Reshape to wide. dy = re_78 - re_75
reshape wide re, i(id) j(year)
gen dy = re78 - re75

global X "age agesq educ educsq marr nodegree black hisp re74 re75 u74 u75"

* Step 1: regress dy on X using CONTROLS only
reg dy $X if ever_treated == 0

* Step 2: predict counterfactual dY_hat for treated
predict dy_hat
gen tau_hat = dy - dy_hat if ever_treated == 1

* Step 3: ATT = mean of tau_hat over treated
summarize tau_hat if ever_treated == 1
```

Spec C in R

```
library(dplyr); library(tidyr)

wide <- d |>
  pivot_wider(names_from = year, values_from = re,
              names_prefix = "re_") |>
  mutate(dy = re_78 - re_75)

# Step 1: OR model on controls only
or_model <- lm(dy ~ age + agesq + educ + educsq + marr + nodegree
                + black + hisp + re74 + re75 + u74 + u75,
              data = wide |> filter(ever_treated == 0))

# Step 2: predict counterfactual for treated
treated <- wide |> filter(ever_treated == 1)
treated$tau_hat <- treated$dy - predict(or_model, treated)

# Step 3: ATT
mean(treated$tau_hat)
```

Spec C in Python

```
import statsmodels.formula.api as smf

wide = d.pivot(index="id", columns="year", values="re").reset_index()
wide["dy"] = wide[78] - wide[75]
# (merge ever_treated and the X's back in)

X = ("age + agesq + educ + educsq + marr + nodegree "
     "+ black + hisp + re74 + re75 + u74 + u75")

# Step 1: OR model on controls
or_model = smf.ols(f"dy ~ {X}",
                   data=wide.query("ever_treated == 0")).fit()

# Step 2: predict counterfactual for treated
treated = wide.query("ever_treated == 1").copy()
treated["tau_hat"] = treated["dy"] - or_model.predict(treated)

# Step 3: ATT
print(treated["tau_hat"].mean())
```

Spec C result: $1,263 — closer still

RCT benchmark	$\$1{,}794$
Spec 0 (no covariates)	$\$3{,}522$
Spec A (additive X)	$\$3{,}522$
Spec B (post$\times X$)	$\$1{,}184$
Spec C (saturated OR / regression adjustment)	$\$1{,}263$

Allowing τ to depend on X moves the estimate further toward the truth — but no regression estimator gets all the way. That's why propensity-score matching (Dehejia–Wahba) and DR-DiD (Sant'Anna–Zhao) exist:
they put $\widehat{p}(X)$ in the right place.

What this buys you: conditional PT is credible

LaLonde's lesson, three decades on:

- Without covariates, observational DiD on the non-experimental sample is **wrong**.
- With covariates entered *correctly* (interacted with post, or via matching / IPW / DR), it moves toward the truth.
- Dehejia & Wahba (2002) and Słoczyński (recent) recover $1,794 closely on this same data using propensity-score methods.

Conditional PT is a real, defensible identifying assumption. The empirical evidence is exactly this exercise — and it's why the rest of the afternoon was worth the time.

The lab: run the four covariate-aware estimators (TWFE-X, OR, IPW, DR-DiD) on the same data and see which one gets closest.

Lab: LaLonde under selection (*the four estimators, one truth*)

- **The benchmark.** The experimental NSW sample identifies the ATT by random assignment: $\widehat{\text{ATT}}^{\text{RCT}} \approx \$1{,}700$ **(1978 dollars)**. Compute it first.
- **The non-experimental panel.** NSW treated $+$ a random CPS sample as the control. Selection bias is large and *negative* — naive DiD gives the wrong sign.
- **Run all four estimators**, get an ATT, compare to $1,700:
  - **TWFE-with-X** (interact Post $\times$ X — without the interaction, FE sweeps them out)
  - **OR** (Heckman–Ichimura–Todd 1997)
  - **IPW** (Abadie 2005)
  - **DR-DiD** (Sant'Anna–Zhao 2020): `drdid` with the `all` option
- **Then answer:** which estimators move the ATT in the right direction? Which get closest? When does DR *not* dominate? What does the propensity-score overlap plot show?

Walkthrough: `glasgow/labs/Lalonde-Covariates/README.md`. *Data:* `lalonde_exp_panel.dta` & `lalonde_nonexp_panel.dta` *on Mixtape-Sessions GitHub.*

What we've done this afternoon

1. Conditional PT — the new identifying assumption
2. Outcome regression: model the trend, impute the counterfactual
3. IPW: reweight controls to look like the treated
4. Doubly robust: pscore as a *weight* (not regressor) + outcome model — and why *controlling for* the pscore is the trap
5. Hidden linearity bias (Caetano-Callaway): *same disease*, costume = TWFE-with-controls
6. Compositional change (Hong; Sant'Anna-Xu): when the *units* shift, not X

The unifying lesson: covariates on the additive RHS of an outcome regression impose constraints on τ.
*DR's escape is to use them as **weights**, not regressors.*

Tomorrow

- **Day 3:** when even covariates can't save you. Synthetic control, the frontier methods.
- Plus: when treatment timing differs across units — the staggered-adoption mess (Goodman-Bacon, Callaway-Sant'Anna, Sun-Abraham, BJS, dCdH).

Use the pscore as a **weight,**

not as a **regressor.**

That is the difference between **doubly robust** and **doubly biased.**